

Vlasov-Poisson Equation with H :

$$\partial_t f + v \partial_x f - (E + H) \partial_v f = 0$$

where $\partial_x E = 1 - \rho_f$, $\rho_f(t, x) = \int_{\mathbb{R}} f(t, x, v) dv$.

K-step: $\partial_t K_i = \langle \text{RHS}_K(K(t, x), V^n), V_i^n \rangle_v$

$$= \langle -v \partial_x (K V^n) + (E + H) \partial_v (K V^n), V_i^n \rangle_v$$

$$= \langle -v \sum_{j=1}^J \partial_x K_j V_j^n + (E + H) \sum_{j=1}^J K_j \partial_v V_j^n, V_i^n \rangle_v$$

$$= - \sum_{j=1}^J \langle v V_j^n, V_i^n \rangle_v \partial_x K_j + (E + H) \sum_{j=1}^J \langle \partial_v V_j^n, V_i^n \rangle_v K_j$$

$$K^{n+1} = K^n + \Delta t \partial_t K^n$$

$$K^{n+1} \stackrel{\text{QR}}{=} U^{n+1} S^*$$

S-step: $\partial_t S_{ij}(t) = - \langle \text{RHS}_S(U^{n+1} S V^n), U_i^{n+1} V_j^n \rangle$

$$= - \langle \text{RHS} \left(\sum_{k=1}^J \sum_{l=1}^J U_k^{n+1} S_{kl}^* V_l^n \right), U_i^{n+1} V_j^n \rangle$$

$$= \sum_{k=1}^J \sum_{l=1}^J S_{kl} \langle \partial_x U_k^{n+1}, U_i^{n+1} \rangle_x \langle v V_l^n, V_j^n \rangle_v - \sum_{k=1}^J \sum_{l=1}^J S_{kl} \langle (E + H) U_k^{n+1}, U_i^{n+1} \rangle_x \langle \partial_v V_l^n, V_j^n \rangle_v.$$

$$S^{**} = S^* + \Delta t \partial_t S^n \Rightarrow L^n = V^n S^{**T}$$

L-step: $\partial_t L_i^n = \langle \text{RHS}_L \left(\sum_k U_k^{n+1} L_k^n \right), U_i^{n+1} \rangle_x$

$$= -v \sum_{k=1}^J \langle \partial_x U_k^{n+1}, U_i^{n+1} \rangle_x L_k^n + \sum_{k=1}^J \langle E U_k^{n+1}, U_i^{n+1} \rangle_x \partial_v L_k^n$$

$$L^{n+1} = L^n + \Delta t \partial_t L^n$$

$$L^{n+1} \stackrel{\text{QR}}{=} V^{n+1} S^{n+1T}$$

set $J(H) = \frac{1}{2} \| f(H)(T) - f^{\text{eq}} \|_{L^2(x, v)}^2$

$$J_r(H) = \frac{1}{2} (\| f_r \|_{L^2}^2 - 2 \langle f_r, f^{\text{eq}} \rangle_{x, v} + \| f^{\text{eq}} \|_{L^2}^2)$$

$$= \frac{1}{2} (\| S \|_F^2 - 2 \langle S, \langle f^{\text{eq}}, (U_i(T) V_j(T))_{ij} \rangle \rangle_T + \| f^{\text{eq}} \|_{L^2}^2)$$

$$J(f_r, H, P) = \frac{1}{2} \| f_r(T) - f^{\text{eq}} \|^2 + \int_0^T \langle g, F_r(f_r, H) - \partial_t f_r \rangle dt.$$

$$\delta_P J = 0 \Rightarrow \partial_t f_r = F_r(f_r, H) = P(-v \partial_x f_r + a(f_r, H) \partial_v f_r)$$

$$\delta_{f_r} J = 0 \Rightarrow \text{set } f_r^\epsilon = f_r + \epsilon \eta, \quad \eta|_{t=0} = 0.$$

$$\Rightarrow \delta J = \langle f_r(T) - f^{\text{eq}}, \eta(T) \rangle$$

$$\delta \int_0^T \langle g, F_r(f_r, H) - \partial_t f_r \rangle dt.$$

$$= \delta \int_0^T \langle g, D_{f_r} F_r \eta \rangle - \langle g, \partial_t \eta \rangle dt$$

$$= \delta \int_0^T \langle (D_{f_r} F_r)^* g, \eta \rangle + \langle \partial_t g, \eta \rangle dt - \langle g, \eta \rangle|_T.$$

thus, $P(T) = f_r(T) - f^{\text{eq}}$

$$\partial_t g + (D_{f_r} F_r)^* g = 0$$

Discrete perspective

$$Z^n = (U^n, S^n, V^n), \quad Z^{n+1} = \Phi_n^{KSL}(Z^n, H^n)$$

$$J_n(Z^n) = \frac{1}{2} \| U^n S^n (V^n)^T - f^{\text{eq}} \|_h^2$$

In a forward step:

$$K_*^n = U^n S^n + h F_0^n W_v V^n$$

$$(U^{n+1}, \hat{S}^n) = \text{QR}_{W_{x,+}}(K_*^n)$$

$$\tilde{S}^n = \hat{S}^n - h (U^{n+1})^T W_x F_1^n W_v V^n$$

$$L_*^n = V^n (\tilde{S}^n)^T + h (F_2^n)^T W_x U^{n+1}$$

$$(V^{n+1}, (S^{n+1})^T) = \text{QR}_{W_{v,+}}(L_*^n)$$

$$\begin{aligned} \mathcal{L}_d = & J_h(U^n, S^n, V^n) + \sum_{n=0}^{N-1} 1 < P_K^n, U^n S^n + h F_0^n W_V V^n - K_K^n >_F \quad (J_K) \\ & + < P_S^n, \hat{S}^n - h(U^{n+1})^T W_X F_1^n W_V V^n - \hat{S}^n >_F \quad (J_S) \\ & + < P_L^n, V^n (\hat{S}^n)^T + h(F_2^n)^T W_X U^{n+1} - L_K^n >_F \quad (J_L) \end{aligned}$$

$$\delta \mathcal{L}_d = \delta J_h + \sum_n (\delta J_K^n + \delta J_S^n + \delta J_L^n)$$

$$\begin{aligned} \delta J_K^n &= < P_K^n, \delta U^n S^n + U^n \delta S^n + h \delta F_0^n W_V V^n + h F_0^n W_V \delta V^n - \delta K_K^n > + < \delta P_K^n, \dots > \\ \delta J_S^n &= < P_S^n, \delta \hat{S}^n - h \delta(U^{n+1})^T W_X F_1^n W_V V^n - h(U^{n+1})^T W_X \delta F_1^n W_V V^n \\ &\quad - h(U^{n+1})^T W_X F_1^n W_V \delta V^n - \delta \hat{S}^n > + < \delta P_S^n, \dots > \\ \delta J_L^n &= < P_L^n, \delta V^n (\hat{S}^n)^T + h \delta(F_2^n)^T W_X U^{n+1} + h(F_2^n)^T W_X \delta U^{n+1} - \delta L_K^n > + < \delta P_L^n, \dots > \\ \delta J_h &= < w \delta V^n (S^n)^T, \delta U^n >_F + < w(U^n)^T \delta V^n, \delta S^n >_F + < w \delta U^n S^n, \delta V^n >_F \\ &\quad (G = U^n S^n (V^n)^T - f^{eq} \quad w = \Delta x \Delta V) \end{aligned}$$

δL_K :

L_K related to $\delta L_K^n, \delta V^{n+1}, \delta S^{n+1}$

$$\begin{aligned} \delta L_K^n \mathcal{L}_d^n &= - < P_L^n, \delta L_K^n > \\ &+ < P_K^{n+1} S^{n+1} + g_0 V^{n+1} S^{n+1}^T, \delta U^{n+1} > \\ &+ < U^{n+1} P_K^{n+1} + U^{n+1} g_0 V^{n+1}, \delta S^{n+1} > \\ &+ < h W_V F_0^{n+1} P_K^{n+1} + g_0 U^{n+1}, \delta V^{n+1} > \\ &< V^{n+1}, (S^{n+1})^T > = DR_{W_V, \cdot} (L_K^n) \\ &\text{then } (\delta V^{n+1}, \delta (S^{n+1})^T) = DR_{W_V, \cdot} (L_K^n) [\delta L_K^n] \\ &V = Q(L_K) \Rightarrow \delta V = DQ(L_K) [\delta L_K] \\ &\delta S = (DR(L_K) [\delta L_K])^T \\ &P_L^n = \bar{L}_K^n = DQ(L_K)^* \bar{V}^{n+1} + DR(L_K)^* (\bar{S}^{n+1})^T \end{aligned}$$

$$\begin{aligned} f_0^n &= U^n S^n (V^n)^T \quad F_0^n = \text{RHS}(f_0^n, H) \\ f_1^n &= U^{n+1} \hat{S}^n (V^n)^T \\ f_2^n &= U^{n+1} \hat{S}^n (V^n)^T \\ \delta f_0 &= \delta U S V^T + U \delta S V^T + U S \cdot \delta V^T \\ \delta F_0 &= D_P \text{RHS}(f_0, H) [\delta f_0] \\ (\delta U^{n+1}, \delta \hat{S}^n) &= D_{QR_{W_X}^*} (K_K^n) [\delta K_K^n] \\ (\delta V^{n+1}, \delta (S^{n+1})^T) &= D_{QR_{W_V}^*} (L_K^n) [\delta L_K^n] \end{aligned}$$

$$\begin{aligned} \bar{F}_0 &= h P_K V^T W_V \\ &< P_K, h \delta F_0 W_V V >_F \\ &= h \text{tr} (P_K^T \delta F_0 W_V V) \\ &= < h P_K V^T W_V, \delta F_0 >_F \\ \Rightarrow \bar{F}_0 &= h P_K V^T W_V \\ \bar{F}_1 &= -h W_X U_1 P_S^T W_V \quad \bar{F}_2 = h W_X U_1 P_L^T \\ g_a &= \bar{f}_a := (D_P \text{RHS}(f_a, H))^* \bar{F}_a \\ < \bar{F}_a, \delta F_a > &= < \bar{f}_a, \delta f_a >_{\delta(U S V^T)} \end{aligned}$$

$$\begin{aligned} \mathcal{L}_d &= J_h(U^n, S^n(L_K^{n-1}), V^n(L_K^{n-1})) - < P_L^{n-1}, L_K^{n-1} >_F + \text{others} \\ \delta L_K^n \mathcal{L}_d &= < \frac{\partial J_h}{\partial V^n}, \delta V^n >_F + < \frac{\partial J_h}{\partial S^n}, \delta S^n >_F - < P_L^{n-1}, \delta L_K^{n-1} >_F = < \bar{L}_K^{n-1} - P_L^{n-1}, \delta L_K^{n-1} > \\ p^n &= U^n (L_K^{n-1})^T \Rightarrow P_L^{n-1} = \Delta x \Delta V (f^n - f^{eq})^T U^n \end{aligned}$$

$\delta \tilde{S}$:

$$\delta \tilde{S} \mathcal{L}_S = - < P_S, \delta \tilde{S} >$$

$$\delta \tilde{S} \mathcal{L}_L = < P_L, V \delta \tilde{S}^T > + h < P_L, \delta F_2^T W_X U_1 >$$

$$\begin{aligned} \delta F_2 &= D_P \text{RHS}(f_2, H) [U_1 \delta \tilde{S} V^T] \\ \Rightarrow h < P_L, \delta F_2^T W_X U_1 >_F &= < h W_X U_1 P_L^T, \delta F_2 > \\ &= < h W_X U_1 P_L^T, D_P \text{RHS}(U_1 \delta \tilde{S} V^T) > \\ &= < (D_P \text{RHS})^* [h W_X U_1 P_L^T], U_1 \delta \tilde{S} V^T > \\ &= < g_2, U_1 \delta \tilde{S} V^T > \\ &= < U_1^T g_2 V, \delta \tilde{S} > \end{aligned}$$

$$\Rightarrow \delta \tilde{S} \mathcal{L}_d = - < P_S + P_L^T V + U_1^T g_2 V, \delta \tilde{S} >_F = 0.$$

$$\text{i.e. } P_S = P_L^T V + U_1^T g_2 V$$

δK_K :

$K_K, U^{n+1}, \hat{S}^n$

$$\begin{aligned} \delta K_K^n \mathcal{L}_S^n &= < P_S^n, \delta \hat{S}^n >_F \\ &- h < P_S^n, (\delta U^{n+1})^T W_X F_1^n W_V V^n >_F = < -h W_X F_1^n W_V V^n (P_S^n)^T, \delta U^{n+1} >_F \\ &- h < P_S^n, (U^{n+1})^T W_X \delta F_1^n W_V V^n >_F = < g_1^n, \delta f_1^n >_F = < g_1^n V^n (\hat{S}^n)^T, \delta U^{n+1} > + < (U^{n+1})^T g_1^n V^n, \delta \hat{S}^n >_F \\ \delta K_K^n \mathcal{L}_L^n &= h < P_L^n, (\delta F_2^n)^T W_X U^{n+1} >_F + h < P_L^n, (F_2^n)^T W_X \delta U^{n+1} >_F \\ &= < g_2^n, \delta f_2^n >_F + \dots \\ &= < g_2^n, \delta U^{n+1} \hat{S}^n (V^n)^T >_F + < h W_X F_2^n P_L^n, \delta U^{n+1} >_F \\ &= < h W_X F_2^n P_L^n + g_2^n V^n (\hat{S}^n)^T, \delta U^{n+1} > \end{aligned}$$

$$C_U := \bar{U}^{n+1} - h W_x F_1^n W_x V^n (P_S^n)^T + g_1^n V^n (\hat{S}^n)^T + h W_x F_2^n P_L^n + g_2^n V^n (\hat{S}^n)^T$$

$$\begin{cases} \bar{U}^N = \Delta x \Delta v \sum U^N S^N (V^N)^T - f^{eq} \int V^N (S^N)^T \\ \bar{U}^{n+1} = P_K^{n+1} (S^{n+1})^T + g_0^{n+1} V^{n+1} (S^{n+1})^T \end{cases}$$

$$C_{\hat{S}} = P_S^n + (U^{n+1})^T g^n V^n$$

$$\begin{aligned} \Rightarrow \delta K_* I_d &= - \langle P_K, \delta K_* \rangle_F + \langle C_U, \delta U^{n+1} \rangle_F + \langle C_{\hat{S}}, \delta \hat{S} \rangle_F \\ &\quad \langle C_U, DQ_x [\delta K_*] \rangle_F + \langle C_{\hat{S}}, DR_x [\delta K_*] \rangle_F \\ \Rightarrow P_K &= (DQ)^* C_U + (DR)^* C_{\hat{S}} \end{aligned}$$

$$\delta_H : \quad \delta f_a^n = 0 \quad a=0,1,2$$

$$\delta_H F_a^n = D_{H,RHS} (f_a^n, H) \int \delta H$$

$$\delta_H I_a = \sum_{n=0}^{N-1} \sum_{a=0}^2 \langle \bar{F}_a^n, D_{H,RHS} (f_a^n, H) \int \delta H \rangle_F$$

$$= \langle \sum_{n=0}^{N-1} \sum_{a=0}^2 (D_{H,RHS} (f_a^n, H))^* \bar{F}_a^n, \delta H \rangle$$

$$\Rightarrow \nabla_H J_h = \sum_{n=0}^{N-1} \sum_{a=0}^2 (D_{H,RHS} (f_a^n, H))^* \bar{F}_a^n$$

Initial condition

$$\text{set } G = U^N S^N (V^N)^T - f^{eq}$$

$$\bar{U}^N = W G V^N (S^N)^T$$

$$\bar{S}^N = W (U^N)^T G V^N$$

$$\bar{V}^N = W G^T U^N S^N$$

Inverse QR

$$A = QR \quad Q^T |wI| Q = I$$

$$\Rightarrow \langle \bar{Q}, \delta Q \rangle + \langle \bar{R}, \delta R \rangle = \langle \bar{A}, \delta A \rangle$$

$$\delta A = \delta Q R + Q \delta R$$

$$(\delta Q)^T (wI) Q + Q^T (wI) \delta Q = 0 \quad (\text{反对称})$$

$$\langle \bar{Q}, \delta Q \rangle + \langle \bar{R}, \delta R \rangle = \langle \bar{Q}, \delta A R^{-1} \rangle + \langle \bar{R} R^T - Q^T \bar{Q}, \underbrace{\delta R R^{-1}}_{\delta A^?} \rangle$$

$$Q^T (wI) \delta A R^{-1} = Q^T (wI) \delta Q + \delta R R^{-1}$$

$$I + I : \quad \delta R R^{-1} + (\delta R R^{-1})^T = Q^T (wI) \delta A R^{-1} + (Q^T (wI) \delta A R^{-1})^T$$

$$\text{th } \delta R R^{-1} \text{ is } \Rightarrow \begin{cases} 0 & i=j \\ (Q^T (wI) \delta A R^{-1})_{ii} & i=j \\ (Q^T (wI) \delta A R^{-1})_{ij} & i < j \\ (Q^T (wI) \delta A R^{-1})_{ji} & i > j \end{cases}$$

$$\Rightarrow \langle \bar{R} R^T - Q^T \bar{Q}, \delta R R^{-1} \rangle$$

$$= \langle \text{copy}(\bar{R} R^T - Q^T \bar{Q}), Q^T (wI) \delta A R^{-1} \rangle$$

$$\text{then } \bar{A} = I \bar{Q} + W Q \text{copy}(\bar{R} R^T - Q^T \bar{Q}) \int R^{-1}$$

$$\text{For } (DQ)^* \quad \& \quad (DR)^*$$

$$\bar{A} = I \bar{Q} + W Q \text{copy}(\bar{R} R^T - Q^T \bar{Q}) \int R^{-1}$$

$$\bar{A} = DQ(A)^* \bar{Q} + DQ(A)^* \bar{R}$$

$$\Rightarrow DQ(A)^* [B] = [B - W Q \text{copy}(B^T Q)] \int R^{-1}$$

$$DR(A)^* [C] = W Q \text{copy}(RC^T) \int R^{-1}$$